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Regulatory and Institutional Factors in Hydropower Sustainability: The Case of Nepal

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Abstract:

Hydropower plays a pivotal role in Nepal's energy landscape, offering a renewable and potentially sustainable source of electricity. However, the sustainability of hydropower development in Nepal is increasingly influenced by complex regulatory and institutional dynamics. This paper critically examines the regulatory frameworks and institutional mechanisms governing hydropower in Nepal, assessing their effectiveness in promoting environmentally and socially sustainable practices. Through a qualitative analysis of policy documents, stakeholder interviews, and case studies of major hydropower projects, the study identifies key gaps in policy coherence, institutional coordination, and enforcement capacity. It also explores how political economy factors and bureaucratic fragmentation impact sustainable hydropower governance. The findings suggest that while Nepal has made progress in policy formulation, institutional inefficiencies and regulatory ambiguities continue to hinder the realization of truly sustainable hydropower development. Recommendations are proposed to enhance regulatory transparency, strengthen institutional collaboration, and align national strategies with global sustainability standards.

Keywords

Hydropower sustainability; Nepal; regulatory frameworks; institutional governance; environmental policy; energy development; policy coherence; sustainable energy; political economy; institutional capacity.

I. Introduction

A. Background

Nepal, endowed with abundant water resources and mountainous terrain, has significant potential for hydropower development. With an estimated economically feasible potential of over 40,000 MW, hydropower is central to the country's energy strategy and economic aspirations. However, while hydropower presents a renewable and low-emission energy source, its sustainability is often compromised by inadequate regulatory oversight, fragmented institutional responsibilities, and limited stakeholder engagement. Issues such as environmental degradation, displacement of communities, and social unrest around major projects underscore the need for a more integrated and effective governance framework.

B. Purpose of the Study

The primary purpose of this study is to investigate how regulatory and institutional factors affect the sustainability of hydropower development in Nepal. It seeks to analyze the adequacy of existing laws, policies, and institutional arrangements in ensuring that hydropower projects are not only technically and economically viable but also environmentally sound and socially just. The study also aims to identify key challenges and opportunities within the governance system that influence long-term hydropower sustainability.

C. Scope and Methodology

This research focuses on Nepal's hydropower sector, examining national regulatory frameworks, institutional roles, and the implementation practices that impact sustainability outcomes. The scope includes both large-scale and small-to-medium hydropower projects, with particular attention to environmental impact assessments (EIAs), licensing processes, inter-agency coordination, and compliance mechanisms. Methodologically,

the study employs a qualitative approach, utilizing document analysis, policy review, and semi-structured interviews with government officials, project developers, civil society representatives, and academic experts. Select case studies of hydropower projects are also analyzed to illustrate systemic strengths and weaknesses in real-world applications.

II. Overview of Hydropower in Nepal

A. Hydropower Potential and Current Capacity

Nepal is endowed with a vast hydropower potential, estimated at around 83,000 MW of theoretical capacity, of which approximately 42,000 MW is considered technically and economically feasible. Despite this enormous potential, the country has historically struggled to harness these resources effectively. As of 2024, Nepal’s installed hydropower capacity stands at just over 2,800 MW, reflecting significant underutilization. The development of hydropower has been characterized by uneven growth, with progress often hampered by political instability, financial constraints, and infrastructural limitations.

B. Role in National Development

Hydropower is central to Nepal’s development vision, contributing significantly to energy security, economic growth, and regional integration. It supports rural electrification, reduces dependence on imported fossil fuels, and offers export potential to neighboring countries such as India and China. Additionally, the government has positioned hydropower as a strategic sector for attracting foreign direct investment and fostering public-private partnerships. In recent years, the sector has also gained prominence in Nepal's climate strategy, aligning with commitments to renewable energy transitions under international frameworks such as the Paris Agreement.

C. Key Challenges Faced

Despite its promise, the hydropower sector in Nepal faces numerous challenges. Regulatory uncertainty, bureaucratic delays, and overlapping institutional mandates often result in project inefficiencies and increased costs. Environmental and social concerns, such as inadequate resettlement practices and ecological degradation, continue to undermine the sector's credibility and sustainability. Moreover, limited transmission infrastructure, financing hurdles, and a lack of technical capacity further constrain hydropower development. External factors, such as geopolitical tensions and climate variability, also pose emerging risks to long-term sectoral stability.

III. Regulatory Framework for Hydropower Development

A. Key Laws and Policies

Nepal's hydropower sector is governed by a combination of laws, policies, and strategic plans intended to regulate project development and ensure sustainable practices. The foundational legal instruments include the Electricity Act (1992) and the Water Resources Act (1992), which establish the legal basis for generation, transmission, and distribution, as well as the allocation of water resources. More recent policy developments, such as the Renewable Energy Policy (2006) and the National Energy Strategy (2013), aim to promote clean energy investment and long-term energy security. The Environment Protection Act (2019) and its corresponding Environment Protection Rules (2020) govern environmental impact assessments (EIAs) and mitigation measures. While these laws provide a basic framework, they are often criticized for being outdated and not adequately aligned with contemporary sustainability standards.

B. Licensing and Approval Process

Hydropower projects in Nepal must navigate a multi-tiered licensing and approval system. The Department of Electricity

Development (DoED) is the primary agency responsible for issuing survey licenses and generation licenses. Environmental clearance is overseen by the Ministry of Forests and Environment, which mandates EIAs for larger projects and Initial Environmental Examinations (IEEs) for smaller ones. Additionally, coordination with the Nepal Electricity Authority (NEA) is required for grid connectivity, while local governments may also have jurisdiction over land acquisition and community engagement processes. Although designed to ensure thorough oversight, the process is often lengthy, non-transparent, and subject to bureaucratic inefficiencies and political influence.

C. Gaps and Overlaps

Despite the presence of numerous regulatory instruments, significant gaps and overlaps exist. One major issue is the lack of coherence between different legal provisions, resulting in conflicting mandates between agencies such as the DoED, NEA, and Ministry of Forests and Environment. Overlapping jurisdictions frequently lead to delays and inconsistent decision-making. Moreover, regulatory enforcement remains weak, with limited capacity to monitor compliance or address violations effectively. Inadequate integration of social safeguards and climate resilience considerations also points to a disconnect between policy intent and on-the-ground implementation. This regulatory fragmentation has deterred investment, created uncertainty, and undermined efforts to promote truly sustainable hydropower development.

IV. Institutional Architecture

A. Key Institutions Involved

The hydropower sector in Nepal involves a complex web of institutions at the national, provincial, and local levels. Key actors include:

- Ministry of Energy, Water Resources and Irrigation (MoEWRI): The apex body responsible for formulating

energy policies, plans, and overseeing implementation across the sector.

- Department of Electricity Development (DoED): Functions as the licensing authority for hydropower projects and serves as a technical regulator.

- Nepal Electricity Authority (NEA): A state-owned utility responsible for generation, transmission, and distribution. It also serves as an off-taker for power purchase agreements (PPAs).

- Ministry of Forests and Environment (MoFE): Oversees environmental clearances, including EIAs and IEEs.

- Investment Board Nepal (IBN): Facilitates large-scale hydropower projects (above 500 MW), particularly those involving foreign investment.

- Local Governments: Under Nepal's federal structure, municipalities and rural municipalities play a growing role in local-level approvals, community consultation, and benefit-sharing arrangements.

B. Roles and Coordination Mechanisms

In theory, these institutions are expected to operate in a coordinated fashion through formal procedures such as joint project review committees, inter-ministerial task forces, and memorandum-based cooperation. For example, EIAs are shared between the MoFE and DoED for joint review, and PPAs require approval from both NEA and MoEWRI. However, in practice, inter-agency coordination is often fragmented, with institutions working in silos and frequently duplicating efforts. The federal transition has further complicated coordination, as roles between federal and local bodies remain ambiguously defined, leading to delays and jurisdictional disputes.

C. Capacity and Governance Issues

Institutional capacity remains a major constraint in Nepal's hydropower governance. Many agencies suffer from limited human resources, outdated technical expertise, and inadequate digital systems for project tracking and data management. Regulatory enforcement is weak, with few mechanisms to monitor compliance post-licensing. Corruption, political interference, and frequent leadership changes further erode institutional integrity and consistency. Moreover, limited stakeholder engagement, particularly with affected communities and civil society organizations, reflects a top-down governance model that undermines social legitimacy. These governance shortcomings contribute to project delays, cost overruns, and social conflicts—ultimately threatening the sustainability of the sector.

V. Sustainability Dimensions and Institutional Influence

A. Environmental Sustainability

Environmental sustainability in Nepal's hydropower sector is critically influenced by the effectiveness of regulatory enforcement and institutional commitment to ecological protection. While Environmental Impact Assessments (EIAs) are legally mandated for most hydropower projects, their quality and enforcement are often inconsistent. Inadequate baseline data, superficial assessments, and weak post-approval monitoring contribute to environmental degradation. Projects frequently impact aquatic ecosystems, disrupt sediment flow, and affect biodiversity due to poor siting and design. Institutional fragmentation—particularly between the Department of Electricity Development and the Ministry of Forests and Environment—has led to poor coordination in managing cumulative environmental impacts, especially in river basins with multiple projects. The absence of strategic environmental assessments (SEAs) further undermines efforts to plan hydropower development holistically.

B. Social Sustainability

Hydropower development has triggered significant social challenges in Nepal, including the displacement of communities, inadequate compensation, and conflict over benefit-sharing. Although legal frameworks exist for resettlement and community development, implementation often falls short due to limited institutional accountability and weak stakeholder engagement. Marginalized groups, such as Indigenous peoples and women, are frequently excluded from decision-making processes. Local governments, now constitutionally empowered, have limited capacity and resources to ensure inclusive and fair processes. Social sustainability is further compromised by inconsistent application of Free, Prior, and Informed Consent (FPIC) principles, leading to resistance and legal disputes in some project areas.

C. Economic and Technical Sustainability

From an economic and technical standpoint, hydropower development in Nepal faces several sustainability hurdles. Financial viability is threatened by project delays, fluctuating water flows due to climate change, and limited access to long-term financing mechanisms. The centralized role of the Nepal Electricity Authority in power purchase agreements has also raised concerns about monopsony risk and pricing inflexibility, particularly for private developers. Technically, challenges include aging infrastructure, insufficient grid capacity, and weak integration of climate resilience in project design. Institutions tasked with oversight—such as the NEA and MoEWRI—often lack adaptive planning mechanisms and robust data systems to anticipate and mitigate these risks, reducing long-term project sustainability.

VI. Challenges and Opportunities

A. Major Obstacles in the Current System

Nepal's hydropower sector continues to face systemic challenges that undermine its sustainability and development potential:

- Regulatory Fragmentation: Conflicting mandates and poor coordination among key institutions result in duplicative procedures, project delays, and regulatory uncertainty.

- Weak Enforcement: Environmental and social safeguards are often inadequately enforced, with minimal post-approval monitoring and insufficient accountability mechanisms.

- Institutional Capacity Gaps: Many agencies operate with outdated technical knowledge, limited financial resources, and inadequate staffing, hindering effective oversight and planning.

- Political Interference and Corruption: Decision-making is often influenced by political interests rather than technical criteria, leading to inconsistencies in project approvals and governance failures.

- Limited Stakeholder Engagement: A top-down approach to hydropower planning marginalizes affected communities and civil society, reducing transparency and increasing resistance to projects.

- Infrastructure Bottlenecks: Transmission constraints and lack of grid integration limit the viability of new projects, particularly in remote and high-potential regions.

B. Emerging Opportunities

Despite these challenges, several opportunities exist to improve the regulatory and institutional landscape and promote sustainable hydropower development:

- Federal Restructuring: Nepal's shift to a federal system offers a chance to decentralize energy governance and empower local governments in project oversight and benefit-sharing.

- Policy Reform Momentum: Ongoing efforts to update outdated laws (e.g., proposed revisions to the Electricity Act) can enhance regulatory clarity, investor confidence, and alignment with sustainability goals.

- Regional Power Trade: Increased regional cooperation, especially through cross-border electricity trade with India and China, presents new market opportunities for Nepal's surplus hydropower.

- Digital and Technical Innovations: Adoption of GIS tools, real-time monitoring systems, and integrated basin planning can strengthen institutional capabilities and reduce environmental risks.

- Climate Financing and Green Investments: Growing international interest in financing renewable energy projects provides avenues to mobilize capital for sustainable and climate-resilient hydropower.

- Public-Private Partnerships: Structured and transparent PPP models can attract expertise and investment while sharing risk and fostering innovation.

VII. Conclusion

A. Summary of Key Findings

This study has explored the complex interplay between regulatory and institutional factors shaping the sustainability of hydropower development in Nepal. While the country possesses vast hydropower potential and a growing policy commitment to renewable energy, current governance frameworks fall short in

ensuring environmental integrity, social equity, and economic viability. Key findings highlight significant regulatory fragmentation, weak institutional capacity, and inconsistent enforcement of safeguards. Although several policies and laws exist, overlaps in mandates, poor inter-agency coordination, and limited stakeholder inclusion hinder their effective implementation. Furthermore, the absence of robust monitoring systems and adaptive planning mechanisms weakens the long-term sustainability of hydropower projects.

B. Implications for Sustainable Hydropower Development

For hydropower to contribute meaningfully to Nepal's sustainable development goals, comprehensive governance reforms are essential. Strengthening institutional coordination, updating outdated legal frameworks, and enhancing regulatory transparency must be prioritized. Equally important is the need to integrate environmental and social considerations into all stages of project planning and execution. Empowering local governments, ensuring meaningful community participation, and leveraging opportunities in regional energy trade and climate finance can significantly improve sustainability outcomes. Ultimately, a shift toward a more integrated, inclusive, and accountable governance model is required to align hydropower development with both national priorities and global sustainability standards.

C. Future Research Directions

Future research should delve deeper into comparative analyses of hydropower governance across South Asia to draw lessons applicable to Nepal. There is also a need for empirical studies evaluating the effectiveness of recent policy reforms and decentralized governance under federalism. Further investigation into the socio-environmental impacts of specific hydropower projects, especially in indigenous and ecologically sensitive regions, would provide grounded insights for policymaking. Finally, interdisciplinary research combining

technical, ecological, and socio-political perspectives can support more holistic and evidence-based hydropower planning in the future.

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